

VIJAYA KARTHIK

Generative AI Engineer | Multimodal RAG | Agentic AI | MLOps & Responsible AI

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PROFESSIONAL SUMMARY

Generative AI Engineer who designed, built and shipped a production multimodal Retrieval-Augmented Generation (RAG) platform on AWS, spanning hybrid vector retrieval, agentic tool calling, self-hosted quantized LLM inference and adversarially tested guardrails. Backed by 2,258 automated tests and a 30-metric MLflow-versioned evaluation harness whose CI gates block any merge that regresses retrieval or hallucination safety.

TECHNICAL SKILLS

Generative AI & LLMs: RAG, Agentic AI, Tool Calling, Prompt Engineering, Self-Hosted LLM Inference, llama.cpp, GGUF Quantization (Qwen2.5-14B), Hugging Face Transformers, LangChain

Retrieval & Vector Search: Vector Databases, Qdrant, Hybrid Search (BM25 + Dense), Cross-Encoder Reranking, Dense Embeddings, Maximal Marginal Relevance (MMR), Semantic Chunking

Evaluation, Safety & Responsible AI: RAGAS, DeepEval, MLflow, LLM-as-a-Judge, Hallucination Detection, Red-Teaming, Prompt Injection Defence, Presidio (PII), OWASP LLM Top 10

ML, NLP & Multimodal: PyTorch, TensorFlow / Keras, scikit-learn, CUDA, GPU Inference, Vision-Language Models, Speech Recognition (ASR), OCR (Tesseract, EasyOCR), OpenCV

Backend & Data: Python, FastAPI, REST APIs, Pydantic v2, Async Python (asyncio), MongoDB Atlas, Redis, Pandas, NumPy, JWT, OAuth 2.0, TOTP MFA, Multi-Tenant Isolation

Cloud, MLOps & Observability: AWS (EC2 GPU, Lambda, CloudWatch, IAM, SSM), Docker, Terraform, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, pytest, Git, Linux

PROJECTS

Multimodal Agentic RAG Integrated Knowledge AI Assistant (MAGIK) v1.0.1 | Finance Domain | Mar 2026 - Aug 2026

github.com/vjkarthik98/MULTIMODAL-AGENTIC-RAG-INTEGRATED-KNOWLEDGE-AI-ASSISTANT | Self-hosted, fully open-weight stack

- **Multimodal ingestion:** Engineered seven isolated ingestion pipelines (text, PDF, DOCX, XLSX, image, audio, video), each owning its extraction, chunking, embedding and BM25 layer so a defect in one cannot degrade another.
- **Hybrid retrieval:** Fused BM25 sparse search with dense vector search on Qdrant, re-ranked by a cross-encoder and diversified with MMR; 0.14s median / 0.23s p95 latency, CI-gated on a 56-query gold set.
- **Agentic routing:** Implemented an agentic query router that classifies and dispatches each query (RAG, web, hybrid, or a finance tool call - financial calculator, SEC EDGAR search) through a typed ToolCall / ToolResult contract, held to a 0.917 routing regression floor.
- **Answer verification:** Grounded every claim against retrieved evidence with a self-verifying layer that retries on failure, matching financial figures to source text at 0.5% tolerance with no unit-scale bridging; 0.94 grounding success, 0.86 citation accuracy.
- **Conversational memory:** Layered short-term Redis session state over persistent MongoDB history with automatic summarization to hold multi-turn context inside the token budget.
- **Guardrails & multi-tenancy:** Hardened all 28 pipeline surfaces with a 12-module guardrail layer (prompt injection, jailbreak, Presidio PII redaction, SSRF, rate limiting) against a 109-prompt red-team corpus - 100% injection recall (64/64) at 0.9% false positives - with per-tenant isolation across Qdrant, MongoDB, Redis and BM25 behind JWT, Google OAuth 2.0 and TOTP MFA.
- **AWS deployment & CI/CD:** Deployed and operated the platform on AWS (EC2 GPU, Docker Compose, Caddy TLS, SSM secrets) across 9 GitHub Actions workflows with zero-downtime releases, auto-rollback and idle GPU

scale-to-zero that cut running cost by roughly 100x versus an always-on instance.

- **Reliability & reproducibility:** Wrapped external calls in circuit breakers and bounded retries; pinned all 18 open-weight checkpoints to exact commit SHAs with SHA-256 verification, locked dependencies and Terraform-codified infrastructure, so a fresh box rebuilds the evaluated system.
- **Evaluation & observability:** Built an 11-suite RAG evaluation harness (RAGAS, DeepEval, LLM-as-a-Judge) versioned in MLflow by Git SHA and prompt version, with fabrication rate at 0.062 against a 0.079 CI ceiling; traced with Prometheus, Grafana and OpenTelemetry.

EXPERIENCE

Generative AI Engineer (Self-Directed Practice) | Jun 2025 - Present

- **Mar 2026 - Aug 2026:** Delivered MAGIK to v1.0.1 in under six months across 43 tagged releases - trunk-based Git workflow, pre-commit quality and security gates (Ruff, mypy, Bandit, pip-audit), and CI that blocks any merge on test, security or evaluation regressions.
- **Jun 2025 - Mar 2026:** Full-time foundation in mathematics, Python, machine learning, deep learning, transformer architectures, NLP, MLOps and Responsible AI, culminating in four certificates.

CERTIFICATIONS

- **Google Cloud** - Responsible AI for Developers, Mar 2026
- **Duke University** - Machine Learning Operations (MLOps) Specialization, Mar 2026
- **IBM** - Generative AI Engineering Professional Certificate, Nov 2025
- **Udemy** - Mathematics - Basics to Advanced for Data Science and GenAI, Jul 2025

EDUCATION

UPSC Civil Services Examination - Full-time preparation | 2021 - 2025

Master of Arts (English) - Bishop Heber College, Tiruchirappalli | 2019 - 2021

Bachelor of Arts (English) - Don Bosco College, Dharmapuri | 2016 - 2019